

Kaveen Sheron Jayasinghe

sheronjay.github.io/my-portfolio | github.com/sheronjay | linkedin.com/in/sheronjay

jayasinghesheronj24@gmail.com | +94 72 385 0350

Education

- B.Sc. Engineering (Hons) in Computer Science & Engineering, University of Moratuwa — 3rd Year Undergraduate (2023 – Present).
- General Certificate Exam Advanced Level, physical sciences stream, Sri Lanka. ZScore +2.2905, with 3 Distinction passes (2022).
- General Certificate Exam Ordinary Level, 9 Distinction passes (2019).
- Primary and Secondary education at St. Joseph's College Colombo-10.

Experience

Collaborator, Oppia Foundation (Google-backed) (2026 - Present)

- Act as Training Co-Lead within the CORE (Creators, Operations, Reviewers, and Editors) team.
- Contribute to full-stack feature development using Angular and Python, develop e2e tests with Playwright, mentor and onboard new contributors, and coordinate quarterly project planning.

Full Stack Developer, ictfromabc (2025 - Present)

- Develop and maintain a Laravel, SvelteKit, and Flutter based learning platform serving 55,000+ registered users.
- Build and integrate full-stack features while improving platform reliability by writing backend tests and contributing to CI workflows.

Projects

Prypiatos

[Prypiatos/repositories](#)

- Led infrastructure implementation by provisioning AWS infrastructure for a 3 node self managed k3s Kubernetes cluster using Terraform, including EC2 instances and EBS storage configuration.
- Automated Kubernetes cluster bootstrap with Ansible, covering control plane setup and worker node joining.
- Implemented a GitOps deployment workflow with Argo CD, enabling continuous synchronization between Git managed Kubernetes manifests and the live cluster state.
- Built MQTT based edge device workflows for real time energy data acquisition, local buffering, reconnection handling and reliable device-to-cloud message delivery.

Cross-Lingual Speech & Identity Synthesis System

[sheronjay/CrossLingual-FaceDub](#)

- Developed an end-to-end English-to-Sinhala multimedia processing pipeline in Python, integrating speech processing, machine translation, speaker-preserving voice conversion, and lip synchronization.
- Built using WhisperX for transcription, NLLB-200 for translation, Coqui TTS and a custom voice conversion model to mimic the original speaker's voice, and Wav2Lip for realistic lip-syncing.

Autonomous Drone Platform for IESL-RoboGames

[IESL-RoboGames](#)

- Contributed to a containerized sandbox environment for autonomous drone development using Docker, supporting Pixhawk, ArduPilot, and Raspberry Pi based workflows.
- Engineered isolated multi-user development containers with SSH access, enabling teams to work independently within dedicated user environments on shared infrastructure.
- Designed persistent storage and recovery mechanisms so user data, code, and workspace state remained intact across container restarts or failures, improving reliability of the development setup.

Veri5

[sheronjay/Syncode-Veri5](#)

- Architected a full-stack platform using Next.js, React, TypeScript, Prisma, and PostgreSQL with modular APIs for authentication, test-kit verification, appointments, orders, clinics, and practitioner workflows.

- Implemented a client-side validation pipeline with TensorFlow.js and custom EXIF metadata parsing to run image inference in-browser, detect screenshots or edited uploads, and preserve privacy by keeping raw images off the backend.
- Designed transactional backend workflows with JWT and Google OAuth, hashed one-time status-share tokens, serial-based test-kit verification, and atomic database updates for kit usage, verification state, and appointment slot availability; selected as a Finalist at Devthon 3.0.

Ohana Note

[sheronjay/Ohana-Note](#)

- Designed and implemented a multi service browser game application for secure message sharing, using UUID-keyed records to generate unique recipient-specific game links.
- Built the system using KAPLAY, Node.js, Express, MySQL, and Docker Compose, separating the landing page, backend API, game client, and database into independently deployed services.

N1K0

github.com/N1K0-micromouse

- Advanced micromouse robot powered by ESP32 microcontroller and a custom-designed PCB.
- Features real-time sensor integration and autonomous maze navigation.

Kandypack

[sheronjay/supply-chain-management](#)

- Supply chain management web application built with React, Express, and MySQL.

Open Source Contributions

- Actively contributing to the Oppia Foundation.
- Contributed to WSO2 open source projects during Hacktoberfest 2025.

Leadership & Volunteering

Selection Committee Lead — SLIoT 2026

- Headed participant selection and evaluation for a National level IoT and innovation-focused program, ensuring fair and technical assessment standards.

Program Committee Lead — Hit the Grounds 2025

- Led the planning and execution of an industry-facing competition with participation from 20 leading companies.

Member — AIESEC in Colombo South (2025–2026)

- Contributed to cross-cultural experience programs and internal operations within a global student-led organization.

Interests & Achievements

- Finalist: Devthon 3.0 (2026)
- Finalist: CodeRally (2025)
- Finalist: CyberShield 4.0 (2025)
- Contributor: Asia-Pacific Robot Contest (2025); Winner: Hitachi High-Tech Award
- Sports: Volleyball (All-Island Champion 2019), Cricket (2012–2022)

Technical Skills

Programming: C, C++, JavaScript, Python, SQL, Bash

Web/Frameworks: REST API, React, Angular, Playwright, Laravel, SvelteKit, Express

Cloud & DevOps: Git, Docker, Kubernetes, ArgoCD, Terraform, Ansible, CI/CD, AWS (S3, EC2)

Other Skills

Languages: English (Professional working proficiency), Sinhala (Native proficiency)